Project Title: ExfilTrap

Stateful Detection and Automated Mitigation of Covert DNS Tunneling and Slow-Drip Data Exfiltration

1. The Problem (What is happening?)

DNS (Domain Name System) is the internet's phonebook. Because DNS traffic is essential for browsing, firewalls usually let it pass without inspection. Hackers exploit this trust by using a technique called DNS Tunneling. They take stolen data (like passwords or files), break it into tiny pieces, and hide those pieces inside fake website requests (e.g., randomdata123.evil.com). This allows them to secretly steal data right under the nose of standard security tools.

2. The Base Paper and Its Flaws

A recent IEEE paper (published July 2026) tried to solve this by using a Machine Learning (ML) model to look at the "randomness" (entropy) of DNS queries to guess if they are malicious. However, this approach has two major flaws:

It misses "slow-drip" attacks: If a hacker sends data very slowly over several hours, it doesn't trigger the system's alarms because no single packet looks suspicious enough.

It has high false alarms: Normal network activities (like Windows updates or CDN traffic) also generate weird-looking DNS requests. The ML model gets confused and blocks legitimate traffic.

3. Our Solution (ExfilTrap)

We are building "ExfilTrap," a multi-module software system designed to sit at the edge of a network, intercept live DNS traffic, and actively hunt for data exfiltration. Instead of just looking at packets one-by-one like the base paper, ExfilTrap uses a smarter, three-layer approach:

- **Stateful Session Tracking** — remembers each sender over long periods, so slow "drip" leaks spread over hours still add up and get caught.

- **Dynamic Baseline Engine** — a separate, non-ML statistical layer that learns the network's normal behavior in real time and adjusts sensitivity automatically, cutting false alarms.
- **Random Forest Classifier** — the machine-learning piece inherited from the base paper; scores each query's likelihood of being malicious based on entropy and structure.

- **Payload Decoding (Proof of Guilt)** — for anything the classifier flags, ExfilTrap attempts to actually decode the hidden text. If it decodes into readable stolen data, that's hard proof, not just a probability score.

4. Automated Mitigation

The millisecond an attack is confirmed, ExfilTrap automatically writes firewall rules (iptables/nftables) at the Linux kernel level to instantly drop the hacker's IP address and cut them off the network.